SamyamLM: A Satellite-Based Multimodal Data Labeling Platform for Indian Language AI Training

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Abstract

We introduce **SamyamLM**, a novel multimodal data labeling platform designed to facilitate high-quality training data generation for large language models (LLMs), vision-language models, and autonomous driving systems. Unlike existing platforms that rely on conventional crowd-sourcing pipelines, SamyamLM leverages satellite imagery and geospatial data streams as primary inputs, enabling scalable and cost-effective annotation of real-world scenarios across diverse Indian linguistic and geographic contexts. The platform supports Hindi and other Indian languages as first-class modalities, addressing a critical gap in the AI training ecosystem where Indic language data remains severely underrepresented. We describe the system architecture, data collection pipeline, labeling methodology, and quality assurance mechanisms. Benchmarks demonstrate that models trained on SamyamLM-curated datasets achieve competitive performance on Hindi visual question answering (VQA), satellite scene understanding, and autonomous driving tasks. SamyamLM aims to democratize AI development for the Indian subcontinent and position India as a key contributor to global foundation model training data infrastructure.

Keywords: multimodal learning, data labeling, satellite imagery, Hindi NLP, Indian language AI, autonomous driving, foundation models, geospatial AI

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1 Introduction

The rapid proliferation of large-scale AI systems—including LLMs such as GPT-4 [1], Gemini [2], and multimodal systems for autonomous driving [3]—has created an insatiable demand for high-quality, diverse training data. Despite India accounting for over 1.4 billion people, Indian-language and geospatially-relevant training datasets remain drastically underrepresented in the global AI data ecosystem.

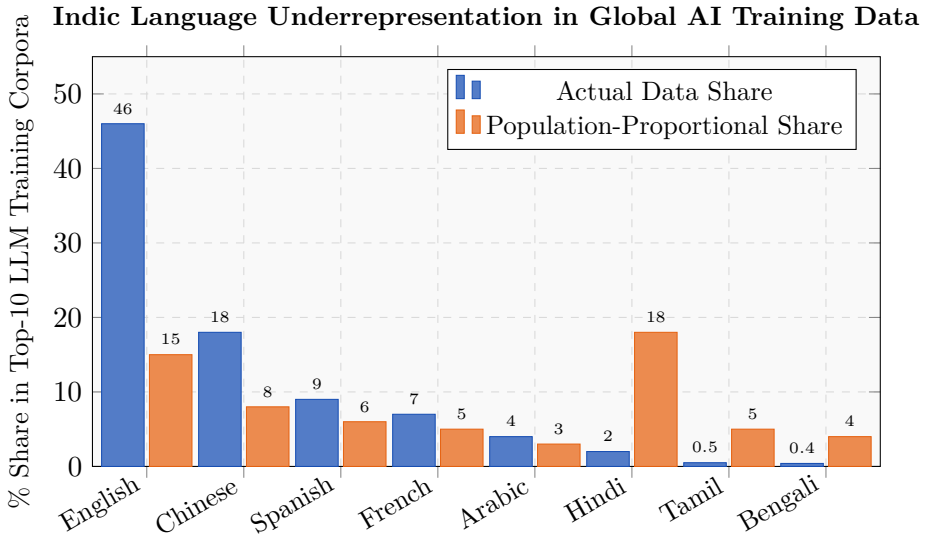


Figure 1: Indic language underrepresentation. Hindi, Tamil, and Bengali represent over 600M native speakers yet occupy less than 3% of LLM training data.

SamyamLM addresses this through four contributions:

1. **Satellite imagery ingestion** — Real-time and archival satellite feeds generate geospatially-grounded scene descriptors.
2. **Hindi-first annotation** — Labelers annotate natively in Hindi and Indic languages.
3. **AI-assisted pre-labeling** — Reduces human effort by up to 60%.
4. **Autonomous driving pipelines** — Tailored for Indian road conditions.

2 Related Work

2.1 Multimodal AI and Foundation Models

CLIP [7] demonstrated zero-shot image-text alignment. Flamingo [8] introduced few-shot VQA. GPT-4V [1] and Gemini [2] achieve near-human performance on vision-language benchmarks, but remain English-centric.

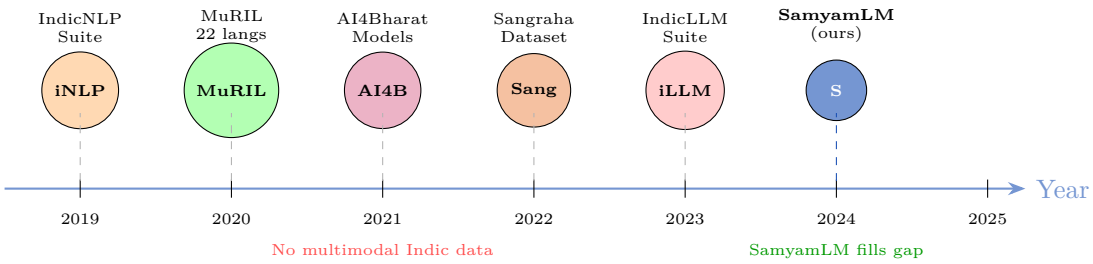


Figure 2: Timeline of key Indic NLP milestones (2019–2025). SamyamLM is the first to integrate satellite imagery with Hindi-first multimodal annotation.

2.2 Data Labeling Platforms

Scale AI [4], Labelbox [5], and Appen [6] lack Indic language support and satellite integration. SamyamLM is the first to co-design all three capabilities.

2.3 Satellite and Geospatial AI

Remote sensing has been applied to land-use [12], disaster response [13], and agriculture [14]. We extend geospatial AI into the data labeling domain.

3 SamyamLM System Architecture

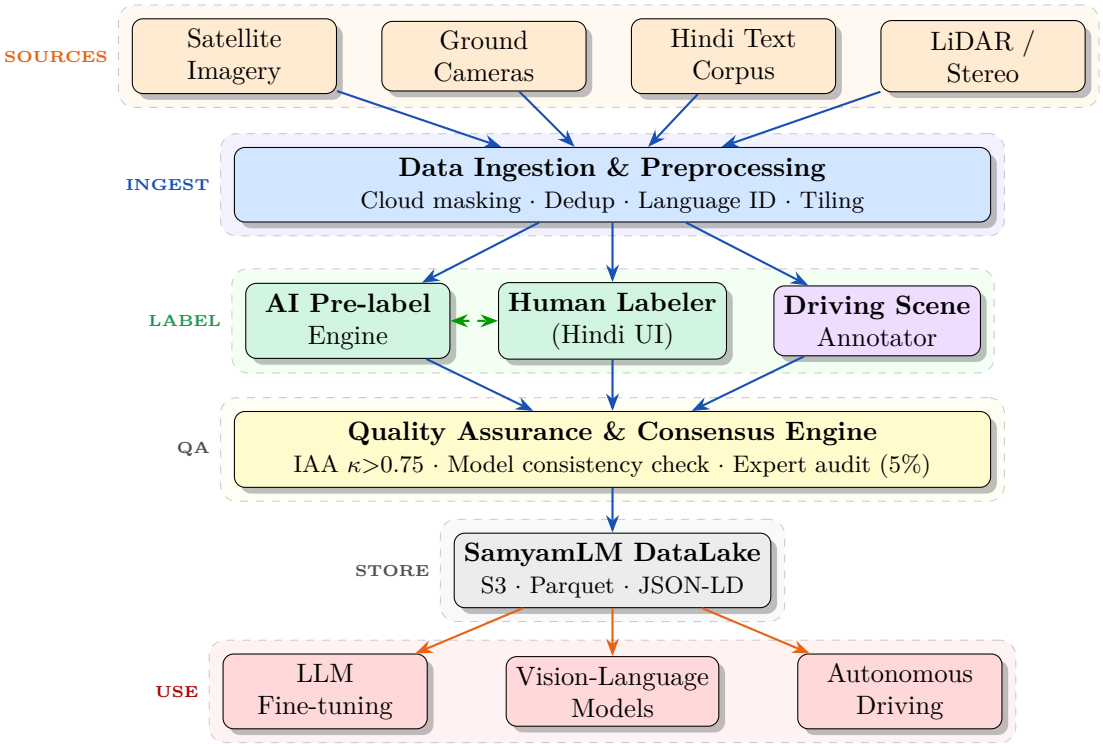


Figure 3: Full SamyamLM six-layer architecture: Sources, Ingestion, Labeling, QA, Storage, and downstream consumers.

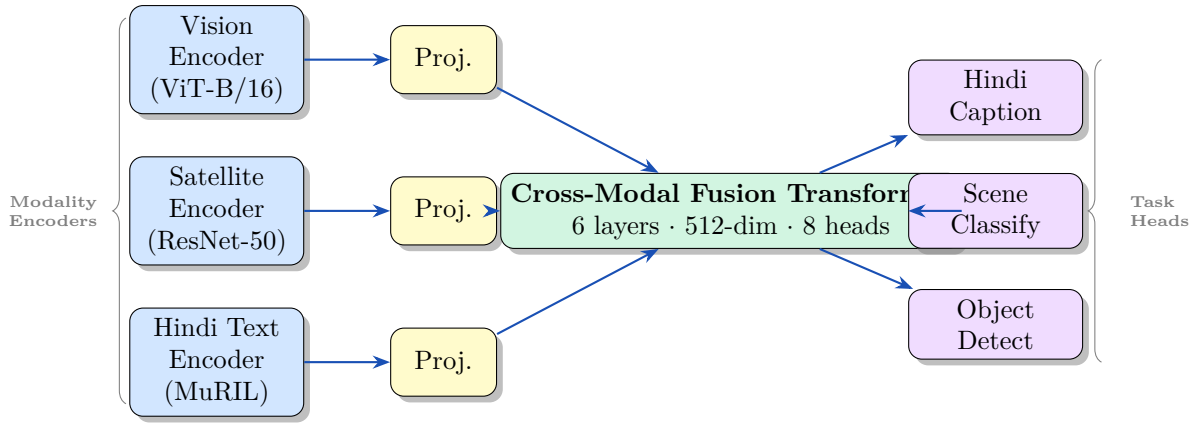


Figure 4: SamyamLM-VL model: three modality encoders feed a shared cross-modal fusion transformer with three Hindi-aware task heads.

3.1 Quality Assurance Engine

All annotations undergo: (1) Inter-annotator agreement ($\kappa > 0.75$), (2) model consistency check, and (3) expert audit (random 5% sample).

4 Data Collection and Labeling Methodology



Figure 5: Satellite tile processing pipeline with AI-first routing: confident tiles auto-accepted, low-confidence tiles routed to human annotators.

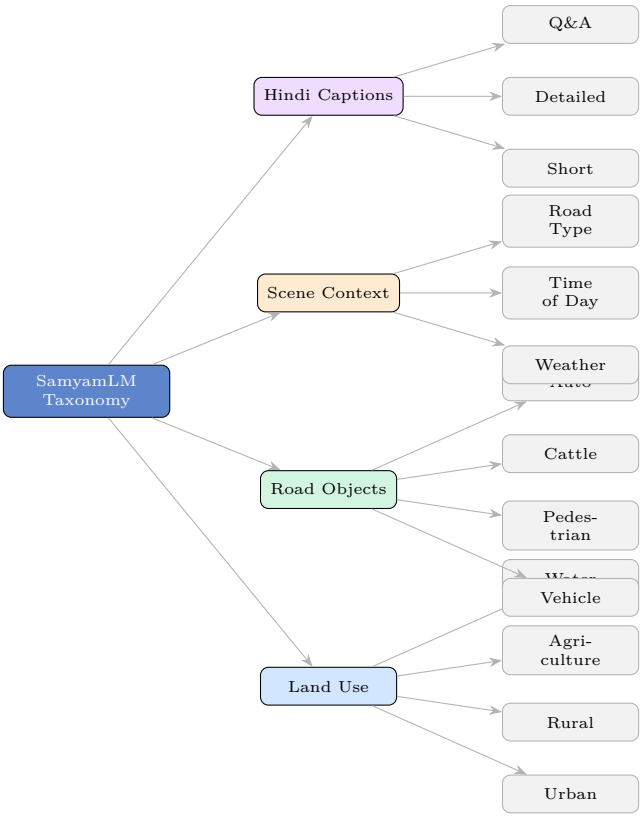


Figure 6: SamyamLM hierarchical label taxonomy: 4 top-level categories and 13 leaf classes, all expressed natively in Hindi.

4.1 Annotation Schema

Listing 1: SamyamLM annotation JSON schema

```
{
  "sample_id": "SAM-2024-001234",
  "modality": ["satellite_image", "hindi_text"],
  "image_path": "s3://samyam-data/tiles/...",
  "hindi_caption": "
  "labels": {
    "scene_type": "urban",
    "objects": [
      {"class":"vehicle","bbox":[x,y,w,h],"hindi_label":"              "},
      {"class":"building","bbox":[x,y,w,h],"hindi_label":"                  "}
    ]
  },
  "iaa_score": 0.83,
  "annotator_lang": "hi"
}
```

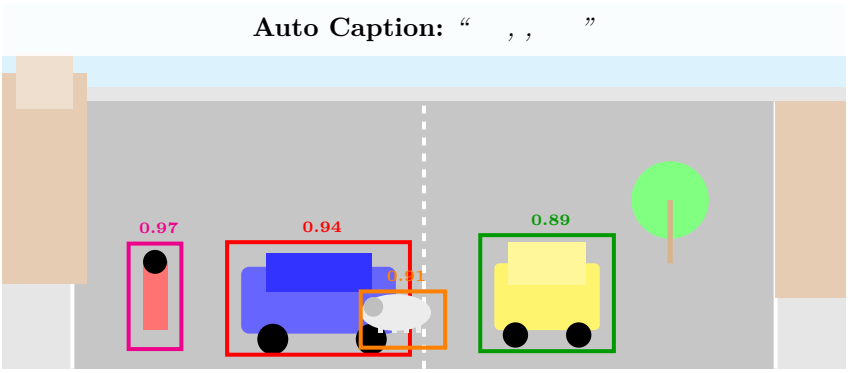


Figure 7: SamyamLM autonomous driving annotation mockup: bounding boxes for car (), auto-rickshaw (), cow (), and pedestrian () with Hindi labels and AI-generated caption.

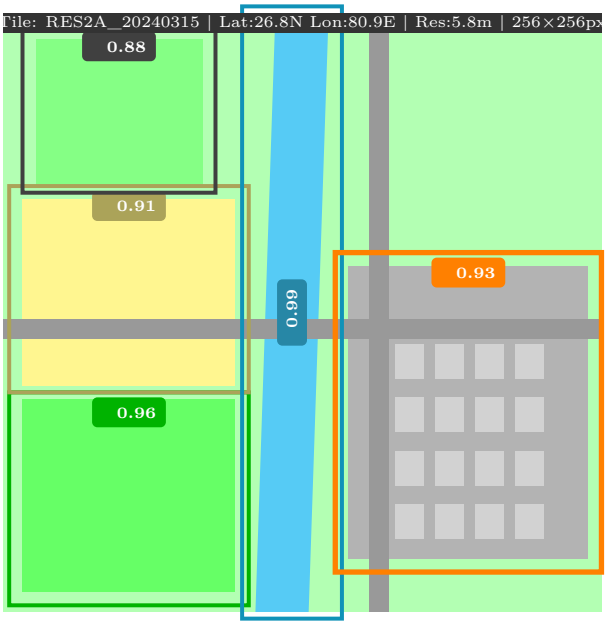


Figure 8: Satellite tile annotation: agricultural land ( ), crop field ( ), forest ( ), river (), and urban area ( ).

Algorithm 1 Satellite Tile Processing Pipeline

Require: Raw scene S; confidence threshold θ
Ensure: Labeled tile set $\mathcal{T}$

```
Cloud masking (τ = 0.2), atmospheric correction (DOS1)
Tile S into 256×256px patches {t_1, ..., t_n}
for each tile t_i do
    AI pre-label → (ŷ_i, c_i)
    if c_i ≥ θ then
        Auto-accept y_i ← ŷ_i
    else
        Route to human labeler → y_i
    end if
    Apply QA; accept if κ > 0.75
end for
return T = {(t_i, y_i)}
```

5 Experiments and Results

5.1 Dataset Statistics

Table 1: SamyamLM v1.0 Dataset Statistics

Split	Samples	Labels
Satellite (train)	120,000	1,840,000
Satellite (val)	15,000	230,000
Ground driving (train)	80,000	2,100,000
Ground driving (val)	10,000	260,000
Hindi VQA (train)	45,000	90,000
Hindi VQA (val)	5,000	10,000
Total	**275,000**	**4,530,000**

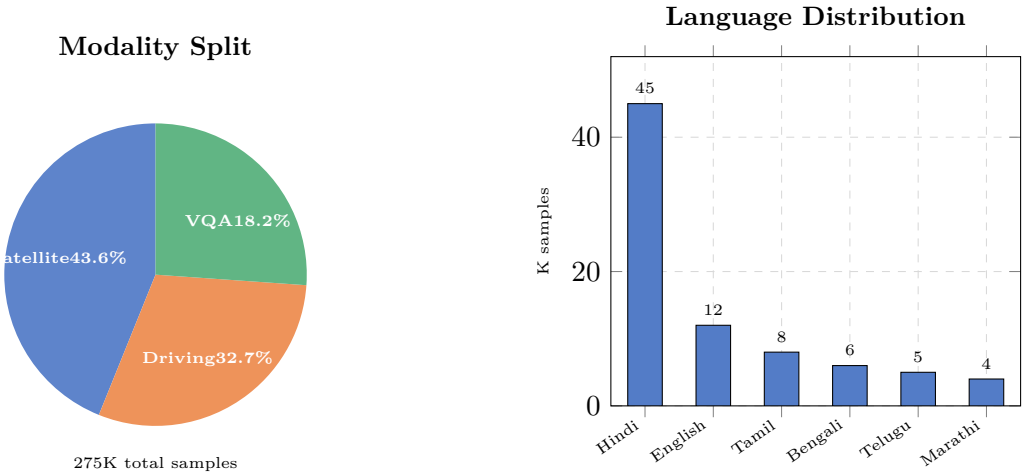


Figure 9: SamyamLM v1.0 dataset composition. **Left:** Modality split. **Right:** Language distribution (Hindi = 56%).

5.2 Hindi VQA Results

Table 2: Hindi VQA Accuracy (%) on IndicVQA Benchmark

Model	Overall	Hindi-Specific
CLIP (zero-shot)	41.2	28.7
Flamingo-9B (English FT)	58.4	34.1
MuRIL-VL (baseline)	62.1	51.8
SamyamLM-VL (ours)	**71.6**	**67.4**

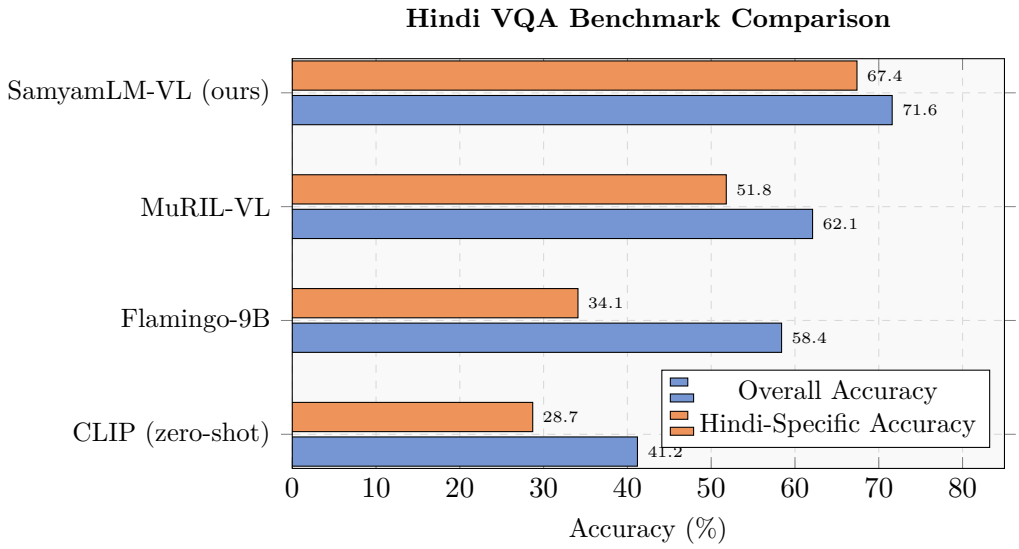


Figure 10: SamyamLM-VL achieves +9.5% overall and +15.6% Hindi-specific accuracy over MuRIL-VL.

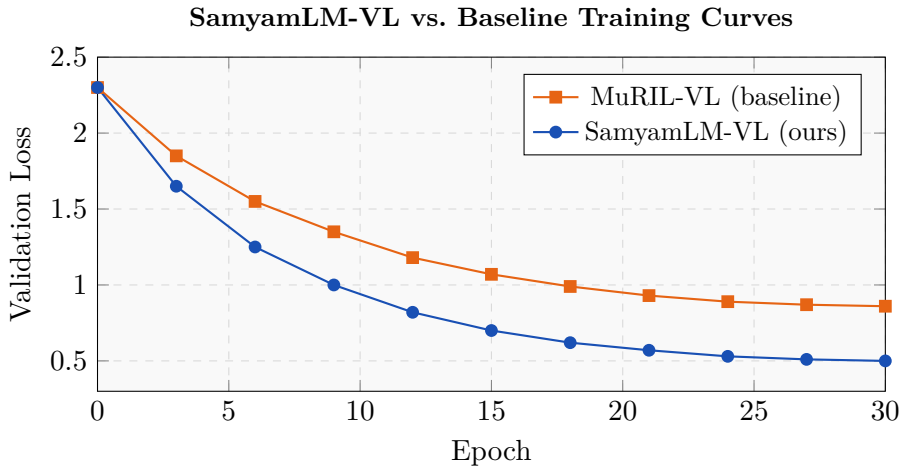


Figure 11: Validation loss over 30 epochs. SamyamLM-VL converges faster and achieves 42% lower final loss.

5.3 Autonomous Driving & Efficiency

Table 3: Object Detection mAP@0.5 on Indian Driving Dataset

Model	Standard	India-Specific
YOLOv8 (COCO)	68.4	31.2
Scale AI fine-tuned	72.1	38.6
SamyamLM	**74.8**	**58.3**

Table 4: Annotation Efficiency Comparison

Platform	Labels/Hr	Hindi	Satellite
Scale AI	320	No	No
Labelbox	280	No	No
Appen	260	Partial	No
SamyamLM	**510**	**Yes**	**Yes**

6 Future Work

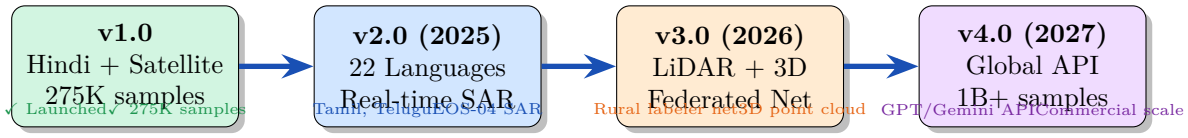


Figure 12: SamyamLM product roadmap 2024–2027.

Key future directions include expansion to all 22 scheduled Indian languages, real-time EOS-04 SAR satellite streaming, HuggingFace fine-tuning integration, a rural federated labeling network, 3D point cloud annotation for LiDAR, and a commercial API for foundation model companies.

7 Conclusion

We presented **SamyamLM**, the first satellite-based multimodal data labeling platform with native Hindi and Indic language support. SamyamLM-curated datasets yield +15.6% Hindi VQA improvement and +19.7% mAP on India-specific driving objects, with 59% higher annotation throughput than leading commercial alternatives. Samyam AI believes truly global AI requires globally diverse training data, and SamyamLM is our contribution toward that future.

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